

CODE SPORT



In this month's column, we discuss how language models are lacking in common sense knowledge and what can be done to address this.

In last month's column, we discussed about how natural language processing (NLP) techniques can be applied in the field of programming and software development. We covered how 'Big Code' data sets enable NLP models to be used for bug detection, comments generation and also how language models can be enhanced for source code modelling. NLP models are also getting applied to a wide number of other fields, which require deep natural language understanding (NLU) and, in certain cases, reasoning capabilities. In this month's column, we will look at how deep NLU requires common sense knowledge capabilities, what is missing in today's large pre-trained language models, and how this can be addressed.

In human-to-human communication, the text that is used for communication does not always explicitly state all the information needed for processing that communication. Consider the following example from the task of natural language inference (NLI). In NLI, you are given a premise and a hypothesis. The task is to predict one of the three classes, namely: (a) entailment – the hypothesis follows from the premise; (b) contradiction – the hypothesis contradicts the premise; and (c) neutral – the hypothesis can neither be inferred nor be contradicted based on the premise. An example is given below.

Premise: The moon was shining brightly and the sun has not risen.

Hypothesis: It was night.

For human beings, this is an example to infer. Since the sun has not yet risen, it must be night. Hence the correct output class is entailment. However, even state-of-the-art neural NLI models get this example wrong. They wrongly predict the output class as contradiction and, in some cases, as neutral whereas human annotators correctly label it as entailment. Why do even sophisticated neural NLP models — even those built on top of BERT family — make this mistake?

This is because human beings have the essential common sense world knowledge that if the sun has not risen, it must be night. However, this is not a common sense fact that NLP models would have learnt, unless it was present in the training data.

You may have the question: Given that large language models like BERT/Roberta, etc, are trained using billions of gigabytes of real world data, how is it possible that they would not have learnt this common sense fact? This happens because of what is known as 'common background knowledge' between all human beings. Because humans have this background knowledge that 'if the sun has not risen, it is night', this fact will never be explicitly communicated in language between two human beings. Since the NLP model will never see this information explicitly in the data it has been trained on, it will not be able to encode this in its internal knowledge representation. Hence, it gets this example wrong.

Let us consider one more example taken from the BERT language model. Given the sentence, "the colour of the dove sitting on the branch is", the BERT language model quite often ends up predicting the next word as red. You can try this example yourself. Because language models are stochastic, you may get different outputs. But after a few tries, you should be able to hit this answer. BERT knows that the answer needs to be a word in the category 'colour'. Based on its distributional knowledge, it knows that colours can be red, blue, grey, green, etc. It will often try to use the most commonly associated word of colour with the word dove. However, it does not know that red doves are rare, so it ends up occasionally choosing the word 'red' when trying to complete the above sentence.

Consider another linguistic task that can be associated with testing for common sense. Here, the task is to fill up the blank with either of the two choices 'always' and 'never'. For example: Given the sentence